

Pull-Muscle and Power Training

You have a finite amount of time and energy to dedicate to training—invest wisely!

While the previous chapter provided explicit details on strengthening your fingers' grip on the rock, the following pages will divulge more than twenty exercises for strengthening the important pulling muscles of the arms and torso. Specifically, you will learn how to train for greater strength, power, and endurance in the muscles that both enable upward movement and help facilitate the static postural positions needed to advance hand and foot engagement.

According to the Principle of Specificity, training gains in strength, power, and local endurance will transfer favorably to climbing only if the exercise is markedly similar in direction of force application, movement patterns, and body position. Based on this principle we can dismiss the common proposition that CrossFit training or bodybuilding exercises are useful for improving rock climbing performance. (These training strategies are certainly legitimate for improving general conditioning and “getting in shape,” but they are not valid as a primary method of training for climbing.) Yes, I do advocate limited use of common barbell exercises—bench press, deadlift, and various dumbbell exercises—for the purpose of antagonist, stabilizer, and posterior chain training, but I do not endorse spending long hours pumping iron or doing vast amounts of other nonspecific training. The bottom line: You have a finite amount of time and energy to dedicate to training, and so you must invest your

time wisely—climbing-specific exercises and actual climbing are where the money is.

In chapter 8 I set forth increasing finger strength as the training-for-climbing prime directive. In this chapter on pull-muscle training, I will likewise stress the importance of increasing maximum pulling strength and power. The standard pull-up, performed with a palms-away grip on a bar, has long been a staple exercise for climbers, and the maximum number of pull-ups performed in a single set is often (incorrectly) viewed as the best measure of pulling strength (see the “Self-Testing Pull-Muscle Strength, Power, and Endurance” sidebar). Upon reaching a modest proficiency at doing pull-ups, however, effective training—and long-term gains in strength—requires more advanced exercises that involve greater resistance than body weight and demand higher rates of force development. Fortunately for you, the pages that follow will detail more than twenty highly specific and effective exercises for increasing your pulling power, lock-off strength, and pull-muscle endurance. Let's get started!

Maximum-Strength Exercises

American free-climbing legend Tony Yaniro famously pointed out that “if you cannot pull through a single hard move, then you have nothing to endure.” So it's with this sentiment that you must resolve to acquire higher and higher levels of maximum strength with each passing training cycle and climbing season. Not only will greater absolute

Self-Testing Pull-Muscle Strength, Power, and Endurance

I'm often asked, "What is the best metric of pull-muscle strength?" Beyond a modest level of climbing fitness (i.e., being able to do only ten or fewer pull-ups), the body weight pull-up is not a good measure of strength; in fact, the maximum number of pull-ups you can do in a single set is a better measure of pull-muscle endurance. For the typical intermediate to advanced climber, then, the best metric of maximum strength is how much weight you can add to your body in doing a pull-up. Here are six pull-muscle tests to track your level of pull-muscle strength, power, and endurance—use them at the beginning and end of a training cycle, or about once quarterly, to track your training gains.

Maximum-strength test #1: What is the most weight you can add and still do a compete pull-up? Clip free weights to the belay loop of your harness, and increase in ten-pound increments to determine the maximum load. The drawback of this test is the large amount of weight needed to accurately test an advanced climber (likely in excess of one hundred pounds).

Maximum-strength test #2: Similar to the previous test except that you're testing to determine the maximum weight you can add in doing five complete pull-ups. (This may take a few workouts to figure out via trial and error.)

strength help you pull harder single moves, but it will also contribute toward higher power output and improved endurance.

As in building maximum finger strength, training to increase pulling and lock-off strength requires high-intensity stimulus that will produce rapid muscular failure. This is a vastly different workout strategy from that described for building

Power test #1: A timed twenty-foot gym rope climb is my preferred test of pulling power. Use a 1.5-inch-thick gym rope and time how long it takes to climb to the top—this usually requires between six and twelve arm pulls, depending on your strength and power. Have a partner time from "first movement" (from a standing position with flat feet) until you touch the clasp at the top of the rope. Advanced climbers can begin from a sitting position.

Power test #2: The time needed to ladder up the 1-3-5-7 rungs on a campus board. Use the largest-size rungs and begin with both hands on the first rung. The stopwatch begins on first movement and ends when both hands match on rung 7.

Power test #3: Beginning with both hands on the lowest large rung of a campus board, pull-up hard (no feet) and slap as high as possible with one hand (but do not engage a high rung with the slapping arm at full extension). Do three slaps with your dominant hand and measure the distance from the top of the starting rung to your high point.

Strength-Endurance test: After a throughout warm-up, do one maximal set of pull-ups—how many full repetitions can you do before failure? Each pull-up should begin with nearly straight arms (but not completely straight with shrugged shoulders) and end with your chin above the bar.

anaerobic endurance (as detailed later in this chapter). Revisit the exercise physiology presented in chapter 5 if you need clarification of the ideal protocols for training maximum strength, power, and endurance. Described below are seven exercises for training maximum strength—employ just two or three of these exercises in any given workout.

Table 9.1 Classification of Pull-Muscle Exercises

	Exercise	Beginner	Intermediate	Advanced
Maximum Strength (anaerobic alactic)	Weighted Pull-Ups	X	X	X
	Square Pull-Ups		X	X
	Steep Wall "Lock-Offs"		X	X
	System Wall "Isolations"	X	X	X
	Uneven-Grip Pull-Ups		X	X
	One-Arm Lock-Offs			X
	One-Arm Pull-Ups			X
Power (anaerobic alactic)	Big Move Bouldering	X	X	X
	Gym Rope Climbing (no feet)		X	X
	Campus Board Laddering (skipping rungs)		X	X
	Boulder Campusing (aka Monkey Business)		X	X
	Clap Pull-Ups			X
	Campus Board Double Dynos			X
Power-Endurance (glycolytic/ anaerobic lactic)	Pull-Up Intervals	X	X	X
	Frenchies	X	X	X
	"Square Dance"		X	X
	Bouldering 4x4s		X	X
	Campus Board "Ladder Laps"		X	X
Local Aerobic Endurance (aerobic/oxidative)	Roped Route Intervals	X	X	X
	Threshold Intervals		X	X
	Concept 2 Rowing Machine	X	X	X
	Endless Rope Machine	X	X	X



WEIGHTED PULL-UPS

Upon being able to do about eight solid body weight pull-ups, you will need to add resistance to continue training maximum strength in the pulling muscles—the best way to do this is simply by adding weight via a weight belt, weighted vest, or by hanging free weights from the belay loop of your harness.

After doing a thorough warm-up, including some basic mobility and stability exercises, the maximum-strength-training protocol is to do three to six sets of weighted pull-ups with enough added weight to make doing five pull-ups challenging. Explode upward as fast as possible with each repetition, but

lower in a slower, more controlled fashion and stop just short of a straight-arm hang position (which must be avoided when training with added weight). Rest for at least three minutes between sets.

If you are new to weighted pull-ups, you will discover that adding just ten or twenty pounds makes for a much more difficult pull-up—you will also discover remarkable gains in pull-up strength in just a few weeks of training! Long term, the amount of weight you need to add may be upward of 50 percent of body weight to make doing five pull-ups a near-maximal exercise. I suggest using a training weight that would allow just seven repetitions if you were doing a single maximal set of weighted pull-ups.



Melissa Le Neve pull-up training with free weights suspended from the belay loop of her harness. CLAUDIA ZEIGLER



SQUARE PULL-UP

This pull-up variation is a bit more climbing specific since it emphasizes lock-off positions and uneven arm force application. Grip a pull-up bar with hands about 50 percent wider than shoulder width, and begin by doing a pull-up to reach the top position. While maintaining the top position, shift to the left by pulling your left hand in against your chest as you push your right hand out to the side as if trying to straighten your arm. Now lower slowly until your left elbow reaches an angle of about 120 degrees, and then immediately shift your torso rightward to a position under the right hand so that the right elbow is now flexed at an angle of about 120 degrees. Pull-up to the top position with your right hand locked off against the right side of your chest and your left arm extended in a nearly straight position. Hold the top position, with chin over the bar, and shift left until your left hand is once again touching your chest—this completes the square pull-up, but keep going!

Do two sets of four to six square pull-ups with a three-minute rest between sets. Do the second set in the opposite direction, and add weight when you find doing two sets of six repetitions to be easy.

How to Pull-Up Train If You Can't Do a Pull-Up?

Pull-ups are the most obvious exercise for a climber to do, but how can you effectively train pull-ups if you can barely eke out one or two repetitions? Here are three excellent assisted pull-up variations that will help you advance to body weight and hypergravity pull-up training in just a few weeks or months.

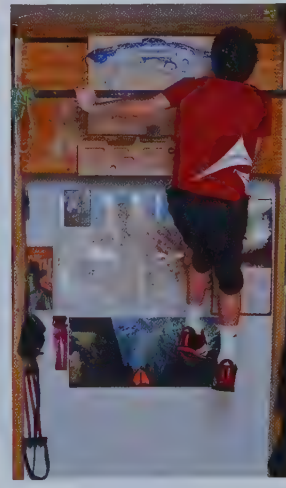
Pull-Up Negatives: Use a chair, box, or other object to step up into the top position of a pull-up with your chin above the bar and hands pulled in tightly near your armpits. Now step off the chair and lower as slowly as possible—subvocalize a five-second count, and try to make the pull-up negative last this long. Upon reaching the bottom position, step back up to the top position and do another slow pull-up negative. Do a total of three sets of five repetitions. Rest for three minutes between sets.

Aided Pull-Ups: The strategy here is to do three sets of five pull-ups with the aid of a training partner or Theraband removing a portion of your body weight. With a Theraband, simply girth hitch one end over the middle of the pull-up bar and then place your feet through the other end. Alternatively, you can have a partner stand behind you and lift around your waist to

remove enough weight so that you can do five pull-ups. Either way, pause for a moment at the top position, and then lower to a two-second count. Do three sets of five repetitions with at least a three-minute rest between sets. Use this exercise three days per week, and soon you'll be doing pull-ups on your own!

Counterweight Pull-Ups: Using a pulley system and counterweights clipped into the belay loop of your harness, do three sets of five to eight pull-ups with a three-minute rest between sets.

Regardless of your chosen technique for assisted pull-up training, it's essential to use proper technique and to do concurrent training of the rotator cuff and scapular stabilizers. Although you can eventually advance to doing pull-ups with your hands at different widths apart, you should initially do them with hands shoulder width apart—and always with a palms-away grip. **Warning:** Avoid bottoming out in the straight-arm position between repetitions, and never hang with straight arms and shrugged shoulders to rest between repetitions.



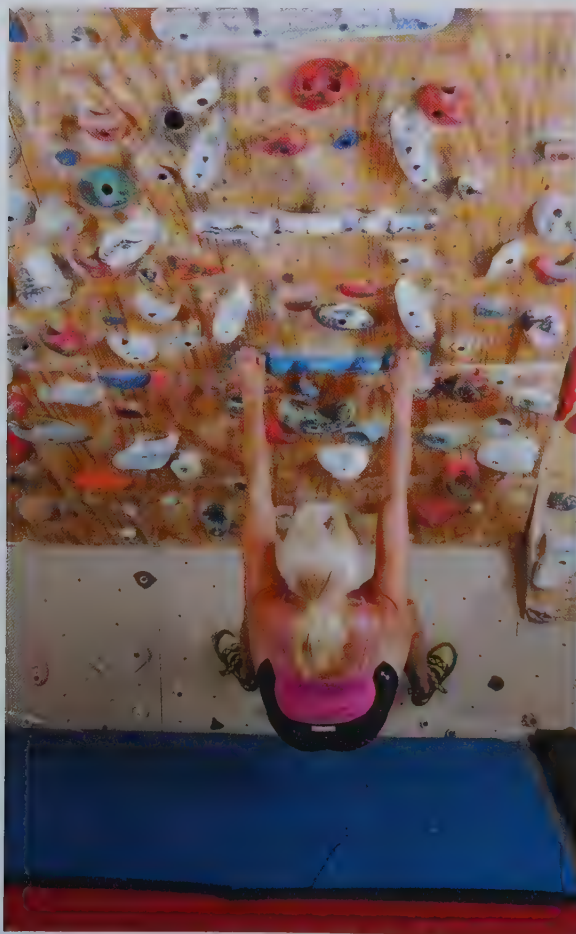
Square pull-up progression.



STEEP WALL “LOCK-OFFS”

This is an excellent exercise for developing lock-off strength and climbing-specific core stiffness. You can do steep wall lock-offs on an overhanging bouldering wall, a HIT Strip or System Wall, or a campus board with foot strips. Begin with matched hands on a pair of good holds (or large campus rung) and both feet on a toe board or footholds, and then pull into a tight lock-off position with one hand and reach as high as possible with the other hand. Touch the wall for two seconds with the free hand while maintaining a solid lock-off with the other—twist

and stiffen your body as needed to maximize your reach. Return to the starting position and immediately pull into a tight lock-off with the other arm. Again, reach as high as possible with the free hand and touch the wall for two seconds. Do not engage any holds with the “touch hand.” Continue for a total of four to six lock-offs on each side. Do two to five sets with a rest of three minutes between sets. Increase difficulty by using a steeper wall or adding a weight belt, but not by using smaller handholds—the focus here is training lock-off strength, not finger flexor strength.



1. Beginning position.



2. Lock-off position.

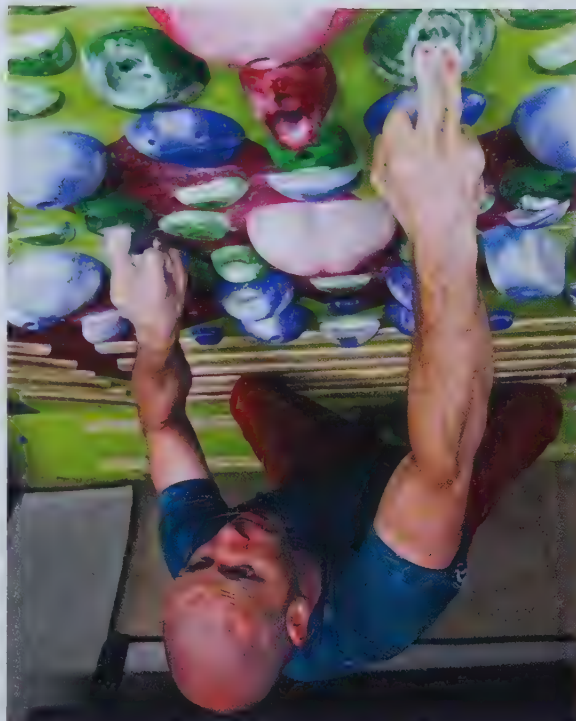


SYSTEM WALL “ISOLATION”

The utility of a System Wall is in being able to isolate and fatigue a specific combination of grip and arm positions while performing actual climbing movements and body positions. While the fingerboard and campus board are better tools for training of finger strength and strength-endurance, the System Wall excels at building pull-muscle strength across a wide range of arm positions. With a well-equipped System Wall, you can isolate a variety of undercling and side-pull positions, pulling and twisting Gaston moves, and compression moves between widely spaced holds. Ultimately, your imagination is the only limitation when it comes to training unique arm and body positions. You can even create a System Wall isolation that mimics a crux move on some outdoor project you’re working—that’s functional training of the highest degree!

Given access to a good System Wall, you may want to reduce time spent bouldering (by ten to twenty minutes) in order to dedicate a little time to isolation training. While you can certainly use small (difficult) handholds to train finger strength on a System Wall, it’s my belief that it’s best to use a fingerboard to train the fingers and a System Wall to train arm positions, body positions, compression, and other specific movements. Each System Wall session should have a focus—what arm and body positions seem to be your weakness on the rock? For many climbers huge functional gains are possible via System Wall Training that isolates long reaches from undercling and side-pull positions, extending reach off a lock-off hold (by pressing it down), twist-locking acutely and deeply for maximum reach, and compressing and slapping up a series of distant holds. Pick two or three different System Wall isolations and do three to five identical repetitions of each, taking only a brief rest in between.

Strive for tight body positions and full extension with each repetition—there’s little benefit to doing incomplete movements with sloppy technique and poor body position. Be sure to perform identical



Kris Hampton training hard on his System Wall.

KRIS HAMPTON COLLECTION

isolations on each side of your body. Increase resistance by doing longer moves or by adding weight (via weight belt) rather than using holds so small and poor that you can’t bear down and fully recruit the core and pulling muscles that you intend to train.



UNEVEN-GRIP PULL-UPS

This excellent exercise develops one-arm and lock-off strength. As the name implies, doing uneven-grip pull-ups requires a setup that offsets one hand twelve to twenty-four inches lower than the other. You can do this by simply looping a sling or knotted rope over a pull-up bar or extending one of a pair of free-floating rings or similar setup.

It’s best to learn the exercise with an opposing grip—do this by standing under the pull-up bar facing longways with the high hand gripping the



1. Starting position (top).

2. Top position—hold for one second (bottom).

bar palm inward (thumb pointing backward) and the low hand gripping the sling with a deep two-finger pocket grip or pinching the knotted rope. From this bottom position initiate upward movement by pulling with both hands; however, focus on pulling hardest with the higher hand. As you ascend past mid-height, begin pushing downward with the low hand to aid the pulling of the high hand. Finish by pulling the bar down along the side of your head to achieve a tight lock-off position. Hold the top position for one second before beginning a controlled descent. Stop short of reaching a straight-arm position and immediately begin the next repetition. Do three to five reps, then dismount and rest for a minute before doing three to five reps with the other hand gripping the bar. Perform two or three sets on each side.

Begin uneven-grip pull-up training with your hands offset by just twelve inches; increase the displacement when you can do more than five reps. Advanced climbers can do this exercise with both hands in a palms-away grip position.



ONE-ARM LOCK-OFFS

The ability to hold a steady one-arm lock-off is vital for hard bouldering and roped climbing. This exercise is obviously very specific to this need, but it does demand a high level of base strength for proper execution. If you cannot hold a solid one-arm lock-off, it would be best to train with weighted pull-ups or uneven-grip pull-ups rather than attempt this exercise. Twice-a-week training with one-arm lock-offs is a perfect gateway exercise to eventually being able to do a one-arm pull-up.

You can do one-arm lock-offs using a pull-up bar or a single free-floating ring. Begin with hands side-by-side and in opposing grips (palms facing each other). The beginning position for each repetition is with hands extended overhead, but not completely straight. Let's train the right arm first: Pull up with both arms to reach a high lock-off position with the bar or ring pulled in tightly near your



One-arm lock-off.

right cheek. Immediately let go with the left hand and hold the static lock-off position as long as possible—when you begin to lose the lock-off, lower as slowly as possible, but do grab back on with the other hand before your elbow reaches an angle of 120 degrees. Immediately commence with another pull-up, this time locking-off in the top position with the other arm. Do two repetitions on each side. Rest for two or three minutes before doing another set or two on each side. **Warning:** Be sure to lower in a slow, controlled manner and never drop forcefully into a fully straight-arm position.



ONE-ARM PULL-UPS

The one-arm pull-up is a benchmark exercise for elite climbers. Amazingly, some of the world's top climbers can do more than five consecutive

one-arm pulls (really hard), and a few ridiculously strong individuals can do a one-arm pull-up while holding twenty pounds (or more) in their free hand! This is obviously not an exercise to rush into training—doing so tempts injury and may be demoralizing—but it is worth playing around with if you've been climbing at a high level for a few years and already possess a high level of base pull-muscle and stabilizer strength. If you can do a few weighted pull-ups with 50 percent of body weight added and a solid one-arm lock-off, then you are likely ready.

First you'll want to learn a one-arm pull-up using a palms-inward grip and by pulling in a neutral position with the bar finishing in a lock-off near the cheek of the pulling arm side. Over time you'll learn to do a one-arm pull-up with the palm facing away and the bar finishing below your chin. The best transition into one-arm pull-up training is by way of a counterweight pulley system; this way you can gradually reduce the amount of counterweight help over the course of many weeks of training. (Alternatively, you can stand in a Thera-band looped over the pull-up bar—stand using the leg opposite the pulling hand.) Either way, do three to five one-repetition sets of aided one-arm pull-ups, using the least amount of counterweight support as possible. Rest for two or three minutes between each one-rep set.

When you sense you are ready to attempt your first body weight one-arm pull-up, I suggest you stand on a box that allows you to begin with just the slightest bend in your pulling arm. Before initiating the one-arm pull-up, it's vital that you tighten your shoulder muscles (engage the rotator cuff and scapular stabilizers), and then think about screwing your forearm inward (supination) as you perform the pull-up motion. Concentrate on pulling the bar down past your cheek (working arm side) and finish in a tight lock-off position—hold a solid finish like a champ, and then take pride in having done your first one-arm pull-up!

Power-Training Exercises

When climbers talk about power, they are typically referring to the need to make quick, strenuous reaches or lunges on steep terrain. This type of movement is the stuff of steep sport climbs and V-hard boulder problems.

Physiologically, your ability to move powerfully is a function of how fast muscle motor units can be called into play and how well they are trained to fire in unison. Effective power training, then, must target the nervous system with fast, dynamic motions that are far different from the strength- and endurance-training exercises covered in this chapter. Inherent to power training are high dynamic force loads, which provide beneficial training stimuli but also threaten the joints and tendons of the fingers, arms, and shoulders. For this reason the following power-training exercises are inappropriate for beginner or recently injured climbers, as well as anyone lacking the maturity and discipline to follow the training and rest guidelines.

Adequate rest between power exercises and workout sessions is also crucial. As a rule you should not engage in more than two power workouts per week. Furthermore, individual workouts should be relatively brief. Training intensity and speed are more important than training volume. In fact, performing a high volume of power-training exercises (or training power in a state of high fatigue) is a prescription for injury. Constantly remind yourself that it's exercise quality, not quantity, that matters most in power training.



BIG-MOVE BOULDER PROBLEMS

If you are an avid boulderer, then you are likely already using this training strategy. The goal is to climb several six- to ten-move boulder problems that involve numerous powerful arm movements between relatively good holds. The ideal boulder will feature mainly positive medium- to large-size holds that will not challenge your grip strength,

and lengthy reaches that demand tight lock-offs, powerful hand movements, and an occasional all-out dynamic move or lunge. When you find the right route, try to send it five times with about a three-minute rest between ascents. Strive to refine your movement and improve climbing economy with each successive ascent—this way you'll learn to climb more accurately and effectively through strenuous movements, despite growing fatigue. If you own a home wall, consider setting a few moderately difficult, big-move boulder problems that you can climb a few times each and every workout.



GYM ROPE CLIMBING

Gym rope and inverted ladder climbing is one of the very best ways to develop awesome upper-body power. Legendary boulderer John Gill used rope climbing as a staple training exercise, and years later John Bachar popularized inverted ladder training among climbers. The Bachar Ladder, as it became known, was a popular training exercise of high-end climbers throughout the 1980s. Since the advent of indoor climbing walls and the campus board, however, rope and ladder climbing have fallen largely out of use. Still, serious climbers would be wise to incorporate some arm-only gym rope climbing into their training program—"campusing" up a 1.5-inch-thick gym rope is, in my opinion, one of the best arm-power-training exercises a climber can do!

After a lengthy warm-up of pull-ups and mild upper-body stretching, begin from either a standing or sit-down (harder) position and grip the rope with both arms at near full extension. Begin with an explosive two-arm pull and then continue upward as fast as possible using quick, crisp arm-over-arm movements. The goal is to maintain smooth, steady upward movement for the duration of the ascent, although it may take some time to develop the necessary arm strength and power. Upon reaching the top, slowly lower back down with controlled arm-over-arm movements—do not drop down in a fast, jerky manner that will



Gym rope training in the Hörst home!

shock-load the elbows and shoulders. If you sense a loss of control, in ascending or descending, immediately clamp down on the rope with your feet rather than risk falling.

Perform three to eight (an elite alactic-power workout) total laps on the rope, always taking at least a three-minute rest between ascents. Optimal adaptations (and gains) to power training come from high-quality, full-speed efforts—think of each ascent as a sprint! Therefore, it's better to do four full-speed sprint ascents, rather than eight low-powered “jogs” up the rope.

Two technical tips: Think about pulling the engaged hand down to meet your upper chest and,

in reaching up, grab the rope with your arm less than fully extended (an elbow angle of between 120 and 150 degrees is good, the latter being much harder).



CAMPUS BOARD “LADDERING”

Chapter 8 presented campus laddering up small holds and without rung skips as an effective method of training contact strength. In the context of training pulling power, however, the effective approach is using larger rungs and longer, more powerful pulls. Most campus boards have at least three different-size rungs (you want to use the largest), with each stack numbered from 1 to 8, and perhaps higher.

The most basic ladder sequence is 1–3–5–7, beginning with both hands on rung #1 and ending with both hands on rung #7 (figure 9.1). As with all power-training exercises, it's essential to move as fast as possible. In the case of a 1–3–5–7 ladder sequence, it should take but a few seconds to make the ascent. Think of this exercise as a sprint—if you need to pause or break sequence, then you're either too tired or too weak to be doing this exercise. Begin by doing just three sets per workout, but build toward a maximum of ten sets over months and years. Rest for at least three minutes between sets, so that you can make a quality effort with each alactic sprint up the board. Elite climbers, however, can train aerobic power by decreasing the rest interval to just thirty seconds between sets.

A more advanced ladder sequence is 1–4–7, although this isn't something you should be in a rush to train. If you regularly fail in attempting the 1–4–7 sequence—or if you engage the #4 or #7 rungs with fully extended arms—then you simply aren't strong enough to do this exercise. **Warning:** Regularly engaging campus rungs with a fully extended arm and an elevated (shrugged) shoulder will quickly lead to shoulder pain and, perhaps, injury. Please heed my advice on this—I can name



Select three- to eight-move routes with medium- to large-size holds that you can engage with an open-hand grip. When you find a boulder with the perfect hold combination, do it three to five times with about a three-minute rest between ascents. Use of good campusing technique is essential—your reaching arm must engage each hold with some observable bend in the elbow and a secured scapula. Have a friend shoot a short video of your boulder campusing, to self-assess the quality of your technique. Favor boulders with fewer moves and moderate reaches to ensure use of good campusing technique.



CLAP PULL-UPS

The clap pull-up is an excellent power-building exercise and a good alternative if you don't have access to a campus board. The huge training benefit of a clap pull-up is that you accelerate all the way to the bar, whereas a standard pull-up requires a deceleration into the bar. This is a difficult exercise, however, and it may take some time to develop the power and timing to release the bar long enough to make the clap.

A good learning progression is, first, to simply release and re-grab the bar without moving your hands at the top of each pull-up. Next, try to release the bar and touch index fingers together before re-grabbing the bar. It helps to lean your head and shoulders back slightly and think about pulling the bar to your chest with each repetition—of course, you must accelerate into the top position and release the bar just before the deadpoint (i.e., the moment of stillness at the top of the movement). Your long-term training goal is two to three sets of three to eight repetitions. Rest for at least three minutes between sets.



CAMPUS BOARD DOUBLE DYNOS

The double dyno exercise, performed on small rungs, was described in chapter 8 as an advanced finger-training method for developing contact strength (page 109). When done using larger rungs and greater flight distance (skipping rungs), the double dyno trains pull-muscle power in addition to contact strength.



Double dynos with rung skips is a most difficult—and potentially injurious—campus training exercise. This is an advanced-only exercise!

Using the largest stack of rungs, begin hanging from one of the middle rungs (say #3), with slightly flexed arms and engaged shoulders. Simultaneously let go with both hands and drop to catch the next lower rung with flexed arms and engaged shoulders. After a brief amortization phase of energy absorption in the pull muscles, explode upward with both hands to catch one rung above the starting rung. This is one full repetition, but don't stop! Without hesitation, drop down to again catch the next lower rung, and then explode back upward to catch two rungs higher. Continue for three or four repetitions. The sequence of rungs engaged, as described above, is 3-2-4-3-5-4-6-5-7. A more advanced double dyno sequence is 3-2-4-2-4-2-4-2-4. Do one to five (advanced) sets with at least a three-minute rest between each set.

As with all power-training exercises, high-speed and high-quality movements are paramount to getting the most out of your double dyno training. The turnaround time in “rebounding” off the low rung must be less than one-half second—if you need to fight to stick the rung or pause to gather yourself, then you're either using too small a rung or you're too tired (or weak) to be doing the exercise. Please review the campus training guidelines (see the “Overview of Campus Board Training” sidebar in chapter 8).

Power-Endurance Exercises

The ability to sustain high power output for thirty seconds to two minutes is largely a function of your anaerobic capacity. More physiologically complex, however, is the capability to prevail through many intermittent bursts of high power output, which depends on the aggregate development of anaerobic power, anaerobic capacity, and aerobic power. (Return to chapter 5 for a review of exercise physiology behind effective training of these energy systems.) Most climbers refer to the above-mentioned traits as power-endurance—an okay term for the ability to sustain powerful locomotion with the

arms, although strength-endurance is a better term in reference to the fatigue resistance of the finger flexors. But I digress.

The focus of this section is training to increase power capacity in the larger pulling muscles, which will provide what you'll perceive to be better power-endurance in climbing. While actual climbing, be it a bouldering 4x4 or a series of short, pumpy climbs, is obviously the most specific training method, there are a few isolation exercises that will directly target the pulling muscles of the arms and torso. Regular use of these exercises—twice weekly is ideal for most climbers—will yield significant gains in this area. Of course, you don't want to train with these grueling exercises too often or without a break every few weeks, as intensive training of the anaerobic lactic (glycolytic) system can often lead to overtraining and performance declines. Consult chapter 10 for guidance on periodizing your training and the need to occasionally taper and break from extensive gym training.



PULL-UP INTERVALS

This is a great exercise that trains the pull muscles to persevere through the fatiguing effects of acidosis (and other by-products of anaerobic glycolysis), as well as strengthens the recuperative influence of the aerobic system (aerobic power). The training goal is to complete twenty pull-up intervals, each exactly one minute in length, composed of a set number of pull-ups performed and a rest period for the time remaining in each interval. For example, if you are doing five pull-ups per interval, it may take ten seconds to complete the five pull-ups, and thus you have the remaining fifty seconds to rest before beginning the next interval (see table 9.2). Use a stopwatch or timing app to stay on a tight schedule for all twenty intervals.

Train using a pull-up bar or the largest holds on a fingerboard. Increase the difficulty by adding one pull-up per interval, rather than using smaller handholds or adding weight. Conversely, reduce the

Table 9.2 Sample Pull-Up Interval Protocol

Interval	# of Pull-ups	Approx. Work/Rest	Interval	# of Pull-ups	Approx. Work/Rest
1	5	10/50 secs.	11	5	10/50 secs.
2	5	10/50 secs.	12	5	10/50 secs.
3	5	10/50 secs.	13	5	10/50 secs.
4	5	10/50 secs.	14	5	10/50 secs.
5	5	10/50 secs.	15	5	10/50 secs.
6	5	10/50 secs.	16	5	10/50 secs.
7	5	10/50 secs.	17	5	10/50 secs.
8	5	10/50 secs.	18	5	10/50 secs.
9	5	10/50 secs.	19	5	10/50 secs.
10	5	10/50 secs.	20	5	10/50 secs.

number of pull-ups per set, as needed, to complete the full twenty-minute interval workout. This is an excellent home training exercise, requiring only a pull-up bar, for individuals without a home wall or easy access to a climbing gym.



FRENCHIES

While it's been more than twenty years since I first coined the name "Frenchies" and popularized this exercise in my book *Flash Training*, this exercise is



1. Lock off at top position.



2. Lock off at 90 degrees.



3. Lock off at 120 degrees.

no less grueling and effective in developing power-endurance in the large pulling muscles.

Begin with a single pull-up (palms away, hands shoulder width apart) and lock off in the top position for a four-second count. Now lower to the bottom (starting position) and pull up to the top again, but this time immediately lower yourself halfway down to an arm angle of 90 degrees. Hold a solid lock-off here for a four-second count, then again return to the bottom position. Immediately crank another pull-up, but this time lower to a lock-off with an arm angle of about 120 degrees—hold this lock-off for four seconds before returning to the bottom position. This sequence of three lock-offs constitutes a single cycle, and it will take about fifteen seconds (assuming it takes one second for each pull-up)—but don't stop yet! Continue on with another cycle (or more) until you can no longer endure the pain and fatigue, or hold a four-second lock-off. Record the number of cycles (or partial cycles) in your training notebook.

Each set of Frenchies should last between thirty seconds (about two cycles) and ninety seconds (six cycles, which is really badass!). If you can't complete two full cycles, stand in a Theraband girth hitched to the pull-up bar or enlist a spotter to remove some body weight by lifting around your waist. Conversely, you can add a ten- or twenty-pound

weight belt once you are able to do five or six full cycles in any given set. Do a total of two or four sets with a rest of at least five minutes between sets.



SQUARE DANCE

This exercise is best performed on a System Wall or campus board with foot strips. Since you'll be training pull-muscle and lock-off endurance, it's important to use large handholds so that your forearms don't pump out. The goal is to "square dance" for between forty-five seconds and ninety seconds.

The exercise is simple, albeit quite boring: Beginning with matched hands on a large low hold, pull-up with both hands and then reach up to grab a large high hold. Next, match hands on the high hold, and then immediately drop the leading hand back to the starting hold. Finish by matching both hands on the starting hold. Hang here for a few seconds, if you like, before repeating the sequence with the other hand leading the way. Keep your feet fixed on the same holds throughout the "square dance." Modulate exercise intensity by varying the distance you reach with each upward movement. Continue in this way for up to two minutes, but step off the wall before reaching complete failure. Do two to four sets with only partial recovery between each set—a work-to-rest ratio of 1:2 or 1:3 is ideal.



Campus board "square dance" using large holds and with feet on large holds or foot strip.



BOULDERING 4X4

In chapter 8 I detailed a highly effective bouldering 4x4 protocol (see table 8.8) for training strength-endurance in the finger flexor muscles. The bouldering 4x4 protocol for developing power-endurance in the pulling muscles is identical, except that you must select boulder problems with better handholds and longer, more powerful arm and body movements. The ideal boulder will demand that your large pulling muscles and core work hard, while not fatiguing the finger flexors to the point of failure. If you fail on one or more of your sixteen boulder ascents, it should be due to fatigue in the pulling muscles (a failed big move, lunge, or lock-off) and not due to failure of the finger flexors (inability to maintain your grip). Therefore, it's best to err on the side of doing routes with handholds that are too large, rather than too small, for the purpose of training power-endurance in the pulling muscles.

In my opinion, it's best to discriminate between strength-endurance training of the finger flexors and power-endurance training of the pulling muscles, instead of training them both at the same time. You can certainly train both in the same workout, but it's best not to do them at the same time. One effective approach is to do a bouldering 4x4 that targets the pulling muscles (as described above) and then, after a rest break of at least twenty minutes, do another bouldering 4x4 that targets the finger flexors (as described in chapter 8). You might conclude your workout with a couple of isolation exercises to further target power-endurance of the pulling muscles (such as a couple sets of Frenchies) and strength-endurance of the finger flexors (as in doing some fingerboard repeaters).



LADDER LAPS

This popular power-endurance exercise involves doing a few laps up and down the largest holds on a campus board. To deeply train the anaerobic lactic

energy system, however, requires that you are strong enough to ladder continuously for thirty to sixty seconds. Since the goal of this exercise is to target the large pulling muscles of the arms and back (not the finger flexors), it's essential to use large handholds—ideally deep rounded edges or large sloper rails—that are easy to grip for the duration of this exercise. Strong climbers will be able to ladder



Perform ladder laps using large holds—the goal is to fatigue the pulling muscles, not the finger flexors.



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TREADWALL M6

lap the campus board with a 1-3-5-7-7-5-3-1 sequence. A somewhat easier approach is to ascend via 1-3-5-7 (skipping rungs) and then descend using every rung (7-6-5-4-3-2-1). Each up-and-down lap will take ten to fifteen seconds, so strive to do between two and four laps per set in order to reach the desired exercise duration of thirty to sixty seconds. Do two to five sets with at least a three-minute rest between sets. Advanced climbers can reduce the rest break to sixty seconds to train aerobic power (recovery ability of CP resynthesis) in addition to anaerobic capacity.

Local Endurance Training

Often overlooked in training-for-climbing programs is local aerobic endurance training. Hopefully the exercise physiology presented in chapter 5 was enough to convince you of the importance of doing some training of the aerobic system, especially if your preference is route climbing.

As detailed in chapter 8's section on aerobic training of finger flexor muscles, the critical training guideline is to climb primarily at a moderate intensity and to avoid developing a deep, painful pump. Ultimately, you want to find the threshold of the anaerobic lactic zone and strive to climb mostly

just below it (in the aerobic zone) while only occasionally crossing into the low end of the anaerobic zone. By keeping your perceived exertion and exercise intensity between 5 and 8.5 on a scale of 1 to 10 (see figure 8.3), you can rest assured that you're training in the proper zone to acquire the desired adaptations.

The specific exercises to aerobically train the pulling muscles are the same as those detailed in chapter 8. Rather than rehash all the fine details here, please consult pages 192–197 for the instruction on local aerobic training via roped climbing intervals and threshold intervals.

Two other nonspecific exercises that offer a solid aerobic workout for the pulling muscles are the Concept 2 Rower and the Endless Rope machine. If you're lucky enough to have access to either one of these machines, then consider adding three sessions per week into your aerobic-training mesocycle and one aerobic maintenance/recovery session per week throughout the remainder of the year. One approach to using these machines is to exercise for three to five minutes at an intensity of 8 out of 10. Do two to four such intervals with a work-to-rest ratio of 2:1 (i.e., two minutes of rest for every four minutes of sustained exercise).

Aerobic endurance training via submaximal Treadwall intervals involving mainly large-hold climbing.

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